GLOBE SCOTT MOTORS · FREE ENGINEERING RESOURCE

DC Motor Gear Ratio Selection Worksheet

For OEM designers and engineers whose motor runs too fast for the driven machine

Fill in the requirement and motor data first, then work through the calculation steps in order. Record every intermediate value so the selection can be reviewed and repeated when the load or supply changes.

Step 1 — Define the requirement

- ☐ Required output speed Nreq in rpm and its tolerance, for example 100 rpm ± 5 %.
- ☐ Running load torque at the output TL in Nm, at worst-case conditions.
- ☐ Load inertia at the output Jload in kg·m² and the required acceleration time.
- ☐ Supply: nominal voltage plus the minimum and maximum voltage at the motor terminals under load.
- ☐ Duty cycle and starts per hour.

Step 2 — Collect motor and gearbox data

- ☐ No-load speed N0 at the actual supply voltage.
- ☐ Stall torque Tstall, or two measured points on the speed–torque line.
- ☐ Rated torque and rated current for continuous duty.
- ☐ Rotor inertia Jm in kg·m², including any brake or encoder.
- ☐ Tolerance on no-load speed from the motor datasheet.
- ☐ Gearbox efficiency at your load for the gear type and number of stages.

Step 3 — Calculate the loaded-speed ratio

- ☐ First guess: i = rated motor speed / Nreq.
- ☐ Motor torque Tm = TL / (i × gearbox efficiency).
- ☐ Loaded motor speed Nm = N0 × (1 − Tm / Tstall).
- ☐ New ratio i = Nm / Nreq; repeat until the ratio changes by less than 1 %.
- ☐ Direct solution: i = [N0 + square root of (N0² − 4 × Nreq × N0 × TL / (efficiency × Tstall))] / (2 × Nreq).
- ☐ If the term under the square root is negative, no ratio can work; choose a larger motor.

Step 4 — Round to a standard ratio and verify

- ☐ List the nearest standard ratios above and below the ideal value.
- ☐ For each, recalculate motor torque, loaded motor speed and output speed; never scale linearly.
- ☐ Prefer the lower ratio (output slightly fast) when voltage or PWM trim is available.
- ☐ Choose the higher ratio only if a slightly slow output is acceptable and torque margin is tight.
- ☐ Confirm continuous motor torque is no more than about 70–90 % of rated torque.
- ☐ Confirm output torque × service factor is within the gearbox rating.

Step 5 — Tolerance and inertia checks

- ☐ Recalculate output speed at minimum and maximum supply voltage and at both ends of the motor speed tolerance.

- [ ] If the spread exceeds the allowed tolerance, plan a commissioning trim, a regulated supply or closed-loop control.
- [ ] Inertia-optimal ratio: i = square root of (Jload / Jm); compare it with the chosen ratio.
- [ ] Acceleration torque at the motor = (Jm + Jload / (i² × efficiency)) × i × output angular acceleration.
- [ ] Running torque plus acceleration torque must stay within the motor's rating and the controller current limit.

Completed example: 100 rpm roller, 9 Nm load, N0 3,450 rpm, Tstall 4.6 Nm, efficiency 0.90

Ratio	Motor torque	Motor speed	Output speed	Verdict
30:1	0.333 Nm	3,200 rpm	106.7 rpm	Select; trim to 94 % duty
32.2:1 (ideal)	0.311 Nm	3,217 rpm	100.0 rpm	Not a standard ratio
36:1	0.278 Nm	3,242 rpm	90.0 rpm	Too slow; cannot recover

Values are for planning. Confirm motor and gearbox data against the manufacturer's datasheets at your actual supply voltage and operating temperature.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.